

Multitask, Multilabel, and Multidomain Learning With Convolutional Networks for Emotion Recognition

Gerard Pons¹ and David Masip², *Senior Member, IEEE*

Abstract—Automated emotion recognition *in the wild* from facial images remains a challenging problem. Although recent advances in deep learning have assumed a significant breakthrough in this topic, strong changes in pose, orientation, and point of view severely harm current approaches. In addition, the acquisition of labeled datasets is costly and the current state-of-the-art deep learning algorithms cannot model all the aforementioned difficulties. In this article, we propose applying a multitask learning loss function to share a common feature representation with other related tasks. Particularly, we show that emotion recognition benefits from jointly learning a model with a detector of facial action units (collective muscle movements). The proposed loss function addresses the problem of learning multiple tasks with heterogeneously labeled data, improving previous multitask approaches. We validate the proposal using three datasets acquired in noncontrolled environments, and an application to predict compound facial emotion expressions.

Index Terms—Convolutional neural networks (CNNs), facial action units (AUs), facial emotion recognition, multitask learning.

I. INTRODUCTION

IMAGES of faces provide relevant information for emotion perception. As humans, we can readily infer a person's emotional state from his or her facial movements [1]. Multiple applications benefit from automated facial emotion recognition, such as human–computer interaction (HCI) [2], student engagement estimation [3], emotionally aware devices [4], or the improvement of expression production in autism spectrum disorder patients [5].

The state of the art in automated facial expression analysis shows excellent performance in the controlled scenario, where images are acquired in studio environments [6]. Nevertheless, the categorization of emotions in uncontrolled environments (also known as *in the wild*) is still an unsolved problem.

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Gerard Pons is with the Distributed and Interactive Systems Group, Centrum Wiskunde and Informatica, 1098 XG Amsterdam, The Netherlands (e-mail: gerard.pons@cwi.nl).

David Masip is with the Department of Computer Science, Universitat Oberta de Catalunya, 08018 Barcelona, Spain (e-mail: dmasip@uoc.edu).

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Besides the strong intraclass variability, facial expression recognition (FER) algorithms *in the wild* must also deal with strong local changes in the illumination conditions, out of plane rotations, large variations in pose and point of view, and low-resolution imaging.

Recent advances in computer vision and particularly in object recognition suggest that new methods based on deep learning can improve the FER task. Convolutional neural networks (CNNs) have represented a relevant breakthrough, especially since the latest improvements on the ImageNet Challenge [7]. However, the amount of available data for the emotion recognition task is small, especially in all the possible configurations of pose, illumination, and resolution. This assumes an inconvenience to exploit the training capacity of these networks, which need large amounts of training data. In this context, the introduction of multitask learning is particularly relevant, as it proved to successfully boost the performance of an individual task with the inclusion of other correlated tasks in the training process [8]–[10]. Thus, tasks with small amounts of data available can benefit from being trained simultaneously with other tasks, sharing a common feature representation and transferring knowledge between different domains. One of the main difficulties with multitask approaches using different datasets is the fact that not all the samples in these different datasets are labeled for all the tasks. For example, one could think of two different datasets of facial images: one for the task of facial recognition (with only labels of person identity) and the other for the task of facial emotion recognition (with only labels of emotion). In order to deal with this problem, classical approaches define different loss functions for each task and train alternatively for the different datasets. That is, when images of the dataset with identity labels are fed in the network, the loss for the identity recognition task is optimized. One could think of the case of a subject whose image is present in both datasets, in one, labeled with his identity, and in the other, labeled with his emotion. When training the image from the identity recognition dataset, only the loss function for the identity recognition task will be considered to update the weights of the network, while this image could also update them according to the emotion recognition task. Our hypothesis is that the use of a unified loss function could improve the performance of multitask learning in datasets of the same typology (i.e., facial images) labeled for different tasks. Hence, in this work, we propose multilabel datasetwise joint loss.

This article makes the following contributions: 1) we formalize a novel datasetwise selective sigmoid cross-entropy loss function to simultaneously train a multitask, multilabel, and multidomain model; 2) we validate that this new proposal outperforms the single-task CNN and the classical multitask approach using different datasets; and 3) we show that the results of joint learning of an unlabeled task are coherently correlated to the labeled task in the case of the emotion recognition problem.

II. RELATED WORK

Automated emotion recognition methods from facial expression analysis in images use a set of six discrete basic emotions that are shared among all cultures [11], namely: *happiness*, *surprise*, *anger*, *sadness*, *fear*, and *disgust*. The production of these emotions in faces depends on specific facial muscle movements. The facial action coding system (FACS) [12] defines a set of action units (AUs) that anatomically group muscle movements. Each expression of emotion can be encoded as a combination of AU activations.

FER is currently a multidisciplinary research field, studied in machine learning, computer vision, cognitive science, psychology, neuroscience, and applied health sciences. The number of computer vision researchers working in the field of facial expression analysis has increased since the early 1990s, and a large number of published works in the topic exist. Depending on the features used for the recognition task, we can distinguish two prevailing methodologies: 1) geometric-based approaches and 2) appearance-based approaches. In the first case, algorithms focus on localizing and tracking specific fiducial facial landmarks, in order to train a classifier based on distances and relative positions of these landmarks. Kotsia and Pitas [13] tracked a set of 119 key points to classify emotions from sequences, and Jeni *et al.* [14] applied a Procrustes transformation on a reduced subset of 117 landmarks for emotion recognition. In appearance-based emotion recognition, a set of features is extracted from pixel images to train a classifier. Classic examples are the Gabor wavelets [2], histogram of oriented gradients (HOGs) [15], and local binary patterns (LBPs) [16]. More complex models use the Gaussian processes [17], hybrid methods using mid-level features [18] or patches from specific key locations [19]. More recently, deep learning methods have outperformed most state-of-the-art algorithms in the literature. Liu *et al.* [20] applied deep learning to a geometric model of facial regions for facial expression analysis and Lu *et al.* [21] used a CNN on facial appearance. Recent models obtain improved results focusing on FER *in the wild* by using committees of several CNN classifiers [22], [23]. Recent surveys on facial expression analysis can be found at [24]–[26].

Despite the current improvements, emotion recognition *in the wild* remains an open problem for the computer vision community. Results from the Emotion Recognition in the Wild (EmotiW 2015 Challenge) reveal modest accuracies [27]. Recently, the first edition of the EmotionNet Challenge (2017) concluded that nonfrontal faces still pose major difficulties

to automated algorithms, and recognition rates decrease as a function of pitch and yaw rotations [28].

Deep learning methods benefit from large training datasets, but the labeling process of facial expression data requires expert FACS coders and is a costly task. Therefore, there are not many datasets available and they are not large enough to exploit the capacities of deep learning. In this article, we conjecture that emotion recognition can be improved if we transfer knowledge from other related tasks. Particularly, we focus on multitask learning. Multitask learning was first studied by Caruana in [29], where he proposed to jointly learn parallel tasks sharing a common representation, and transferring part of the knowledge learned to solve one task to improve the learning of the other related tasks. A detailed analysis of the topic is carried out in some recent surveys [30], [31].

Ranjan *et al.* [32] proposed a multitask approach for face detection, landmark localization, pose estimation, and gender recognition. This method exploits the benefits of multitask learning to improve the performance of each individual task. However, they used a single source of data, hence all the images must be labeled with all the tasks involved.

The closest work to this article was published by Fourure *et al.* [33]. They proposed a multitask CNN for semantic segmentation of outdoor images in order to learn different tasks using images from different datasets. They used a selective soft-max cross-entropy with the objective of not penalizing the training of a task when feeding the model with images annotated for another task. However, this approach does not learn from similar labels. For instance, when training a dataset A with a class labeled as *Grass*, and dataset B with a class labeled as *Vegetation*, the method would predict the label corresponding to the dataset where the input image belonged to (if the image is from dataset A will predict it as *Grass*). In this article, we hypothesize that general accuracies will improve if we learn from both labels at the same time, and take full benefit from training related tasks using different nonhomogeneous datasets.

III. SELECTIVE JOINT MULTITASK LOSS (SJMT)

We propose a novel multilabel loss function that can be integrated into CNNs to improve the training of the emotion recognition task incorporating complementary tasks and data from different sources. In this article, we focus on the multitask training of emotion recognition and facial AUs, but this proposal can be extended to other multilabel problems.

Emotion recognition is a problem where the model classifies the input image as one of the seven emotion classes [11] (including the neutral class). Therefore, when addressing this task using a CNN, the computation of the output probabilities is performed by using the soft-max function in the values of the last layer. This setting is not valid in our multitask approach, since AU recognition is a multilabel problem. The classical setting for multilabel approaches is the use of the sigmoid function for the calculation of the output probabilities. However, using images from different sources and for different tasks implies having partially unlabeled data when performing the multitask training, that is, not all data are labeled for

all the tasks. Thus, this approach is not valid when dealing with images from different datasets since the sigmoid function penalizes the absence of labels in the related task. For example, the *Happy* emotion is generated with the activation of the facial muscles corresponding to the AU6 and AU12. Hence, optimizing the *Happy* class using samples with no AU annotations would penalize the AU6 and AU12 classes provoking a non-desired effect since these AUs are intrinsically present when the emotion *Happy* occurs. We thus propose the selective joint multitask (SJMT) approach which defines a novel dataset-wise selective sigmoid cross-entropy loss function [defined in (1)] to address multitask, multilabel, and multidomain problems.

Given a set of images from K different datasets, with a corresponding set of label spaces $\mathcal{L}$, where each of the label spaces $\mathcal{L}^k$ are different, we define y_j^k the actual label of the j th class in the space $\mathcal{L}^k$. The aim is to learn a nonlinear mapping presented as a CNN which minimizes a cross-entropy loss function for each sample, defined as follows:

$$E(\hat{y}, y, k) = \frac{-1}{N} \sum_{j \in \mathcal{L}^k} \left[y_j^k \log \hat{y}_j + (1 - y_j^k) \log (1 - \hat{y}_j) \right] \quad (1)$$

where N is the number of classes. The term $\hat{y}_j$ is a sigmoid function defined as

$$\hat{y}_j = \frac{1}{1 + e^{-\hat{p}_j}} \quad (2)$$

where $\hat{p}_j$ is the output of the last layer for the j th class of the CNN defined as $(Wh + b)$, W and b are, respectively, the weights and bias of the layer and h is the hidden representation of the last layer.

Even though this cross-entropy function could be incorporated in any CNN architecture, in this article, we used two of the most used architectures in object recognition, VGG-16 [34] and Resnet-50 [35].

In order to visualize the main differences between the strategies compared in this article, Fig. 1 shows a diagram of each approach. First, Fig. 1(a) shows the approach where individual CNN is tailored to each task and dataset. Second, Fig. 1(b) shows the classical multitask approach where the filters of the network are shared during training among the different tasks, but each task has a specific cross-entropy loss function and output layer, and the training is performed alternatively feeding the model with batches of images from each dataset. Finally, Fig. 1(c) shows the approach proposed in this article, where the entire network is shared among the different tasks and images from different datasets are fed indistinctly, thanks to the selective cross-entropy function. Thus, there is only one cross-entropy function and one output layer of size $\sum_{k=1}^K |\mathcal{L}^k|$. During the backpropagation process in the training of the model, the sigmoid selective cross-entropy function updates the weights considering the samples with a corresponding label. This also allows training samples from different datasets in the same batch.

IV. EXPERIMENTAL RESULTS

In this article, we used the following datasets for the experimental validation.

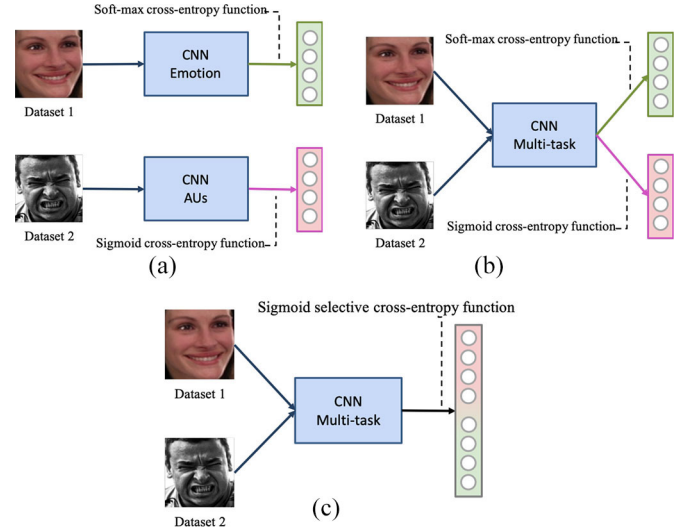


Fig. 1. Different CNN strategies used in this article. (a) Traditional approach with different CNN for different tasks. (b) Classical multitask approach with different output layers and loss functions for each individual task. (c) Proposed multitask approach with a single output layer and the selective sigmoid cross-entropy function.

- 1) *SFEW2.0 dataset* [27], released for a competition in the 3rd Emotion Recognition In the Wild 2015 (EmotiW2015) challenge. The dataset was created by extracting frames from film clips with emotional content to obtain images in close-to-real world conditions. The images are labeled with seven expressions (angry, disgust, fear, happy, sad, surprise, and neutral). The dataset consists of 958 images for training, 436 for validation, and 372 for testing. Since the labels of the test set are not provided, the validation set was used as test set for the experiments in this work. The dataset also provides the same data after being processed with a face alignment algorithm.
- 2) *Oulu-CASIA database* [36], consists of posed videos from 80 subjects in the age group of 23–58 years. The videos are labeled with six emotions (angry, disgust, fear, happy, sad, and surprise). In addition, in this article, the first frame of one video of the subject is used and labeled as neutral. All expressions were recorded in three different illumination conditions: a) strong; b) dark; and c) weak. In the strong category, appropriate lighting is used. In the weak category, only the computer display is on, whereas, the dark category means near darkness. For each illumination category, 480 video sequences were recorded. In order to obtain comparable results with other methods of the state-of-the-art, in this article we used only images with strong illumination. We conducted a ten-fold cross-validation experiment, with subject independent splitting.
- 3) *EmotionNet dataset* [37], released for the EmotionNet Challenge 2017 [28]. The dataset consists of 25 000 images of facial expressions with manual AU annotations. These annotations were given by expert human coders and cross-referenced for verification of accuracy. The images were downloaded from the Internet,

and the authors eliminated repeated and highly similar images. From these images, a subset of 2000 images was annotated with the seven emotion expressions and nine compound emotions made by the combination of two basic emotions, that is, *happily surprised*. These 2000 images were removed from the dataset and used only for the compound emotion recognition task detailed in Section IV-D. From the other 23 000 images, we generated the test set randomly selecting 1000 images (the same for all the experiments) because the official testing set is private and only available for participants of the challenge.

For Resnet [35], we used a training batch size of 32 images of size 224×224 , learning rate of 0.001, and batch normalization to prevent overfitting. For VGG [34], we used a training batch size of 32 images, and a learning rate of 0.0001. In both networks, we used the early stopping criterion to finalize the learning: every time a model outperformed the best previous model in the training, this model was saved. This evaluation was performed leaving out 100 images randomly selected from the training set, and no testing data was used for the parameter adjusting process. In case the model did not improve the results in the following 10 epochs, it was considered that the model was overfitting the training data and hence the training process was stopped.

All the networks trained in this article were pretrained first on a recent dataset for face verification, called VGG-Faces [38]. Then, they were also pretrained on the FER 2013 dataset [39], which consists of 28 709 examples for training, 3589 for validation and 3589 for testing. All the experiments in this work were developed using TensorFlow [40] on NVIDIA GeForce GTX Titan GPU.

With regard to the detection of AUs and recognition of emotion categories, the evaluation measure used was accuracy. Accuracy a measures the number of correctly classified examples, indicating whether our algorithm can discriminate between sample images where a certain AU/emotion is present. Besides, for the detection of AUs, the value of sensitivity is also provided. Sensitivity s measures the proportion of actual positives that are correctly identified as such. They are, respectively, defined in (3) and (4), where i specifies the class, that is, AU_i or the i th emotion category, TP_i (true positives) are correctly identified test instances of class i , TN_i (true negatives) are test images correctly labeled as not belonging to class i , FN_i (false negatives) are test images incorrectly labeled as not belonging to class i and N is the total number of test images

$$a_i = \frac{TP_i + TN_i}{N} \cdot 100 \quad (3)$$

$$s_i = \frac{TP_i}{TP_i + FN_i} \quad (4)$$

Note that the outputs of the sigmoid function are continuous values in the range $[0, 1]$. During the training phase, these values are passed directly to the cross-entropy loss function. However, to calculate the aforementioned metrics in the testing phase, a threshold of 0.5 is applied to transform the continuous values to 0 or 1.

TABLE I
RESULTS OF THE TWO NETWORK ARCHITECTURES USED IN THIS ARTICLE FOR THE EMOTION AND AU RECOGNITION TASK IN DIFFERENT DATASETS

Network	Emotion (accuracy)		AUs (acc. - sensitivity)
	SFEW	Oulu-CASIA	EmotioNet
VGG-16	43.6%	76.8%	85.5% - 0.773
Resnet-50	43.8%	76.3%	86.6% - 0.799

A. Emotion and AU Recognition With Individual Networks

Before validating the proposed approach in this article we validated the performance of each task trained with dedicated individual networks for such a task and dataset. Thus, we trained specific networks for emotion recognition [see Fig. 1(a)] with the SFEW and Oulu-CASIA datasets, and for AU detection with the EmotioNet dataset. Since SFEW and Oulu-CASIA provide samples only labeled with one emotion class, the soft-max cross-entropy function is used. For the EmotioNet, the sigmoid cross-entropy function is used, because multiple AUs can be activated for each sample. Results for the Oulu-CASIA dataset are in terms of average classification accuracy of the ten-fold cross-validation.

Table I shows the accuracy obtained for each CNN for the different tasks and datasets. Note that the images were not preprocessed, specifically to enhance the performance in one dataset or another, and the same parameters were used in all the datasets for a fair comparison. Hence, the most noticeable fact is the similar results obtained by both VGG-16 and Resnet-50. These results will be used as a baseline for further comparisons.

B. Multitask Learning for Emotion and AU Recognition

In this section, we validated the hypothesis that the proposed SJMT method [see Fig. 1(c)] outperforms the individual networks compared in the previous section as well as the classical multitask approach [see Fig. 1(b)]. For this experiment, we jointly trained the multitask approaches for emotion and AU recognition. First, we evaluated the approaches with SFEW and EmotioNet datasets. The obtained results are summarized in Table II. For a clear overview, the results of the individual networks for those datasets are also included in the table.

As expected, both multitask approaches outperformed the single networks in the detection of emotions and AUs. Comparing the SJMT and the classical multitask approach, the proposed method outperformed the classical approach in all experiments, achieving an accuracy of 54.8% in emotion recognition and 89.0% in AU detection with Resnet-50. In terms of sensitivity in the AU detection task, both multitask approaches performed worse than the individual networks. The benefit of multitask training in terms of the smaller dataset (SFEW) seems to be detrimental in the training of the bigger dataset (EmotioNet) in terms of sensitivity. In this case, an individual network training only with the bigger dataset obtained better results.

TABLE II

RESULTS OF MULTITASK LEARNING FOR THE EMOTION AND AU RECOGNITION TASKS FOR THE SFEW AND EMOTIOnET DATASETS. *Multitask* REFERS TO THE CLASSICAL APPROACH [SEE FIG. 1(B)] AND *SJMT* REFERS TO THE SJMT APPROACH [SEE FIG. 1(C)]

Method	Network	Emotion	AUs
		(accuracy)	(acc - sensitivity)
		SFEW	EmotioNet
Individual	VGG-16	43.6%	85.5% - 0.773
	Resnet-50	43.8%	86.6% - 0.799
Multi-task	VGG-16	47.8%	87.8% - 0.726
	Resnet-50	47.6%	87.2% - 0.762
SJMT	VGG-16	48.9%	87.8% - 0.700
	Resnet-50	54.8%	89.0% - 0.751

TABLE III

COMPARISON WITH STATE-OF-THE-ART SOLUTIONS ON SFEW DATASET. (◊) INDICATES APPROACHES USING ENSEMBLES OF MULTIPLE NETWORKS. (*) INDICATES THE BEST SINGLE CNN OF THE ENSEMBLE

Method	SFEW
Levi et al. [41]	41.9%
Pons et al. [23] ◊	42.9%
Mollahosseini et al. [42]	47.7%
Ng et al. [43]	48.5%
Yu et al. [44]◊	55.9%
Yu et al. [44]*	52.3%
Kim et al. [45]◊	52.8%
Kim et al. [45]*	52.5%
Ding et al. [46]	48.3%
SJMT	54.8%

TABLE IV

RESULTS OF MULTITASK LEARNING FOR THE EMOTION AND AU RECOGNITION TASKS FOR THE OULU-CASIA AND EMOTIOnET DATASETS

Method	Network	Emotion	AUs
		(accuracy)	(acc - sensitivity)
		Oulu-CASIA	EmotioNet
Individual	VGG-16	76.8%	85.5% - 0.773
	Resnet-50	76.3%	86.6% - 0.799
Multi-task	VGG-16	84.3%	88.2% - 0.751
	Resnet-50	78.9%	70.9% - 0.773
SJMT	VGG-16	85.1%	84.8% - 0.708
	Resnet-50	81.0%	81.3% - 0.777

Table III compares the proposed SJMT approach with the state of the art on SFEW. For ensemble models, both ensemble and the best single CNN results are reported, with ◊ denoting the ensemble and * denoting the best single CNN. The proposed SJMT achieved a better performance than existing single networks (+2.3%) and nearly the same performance (−0.9%) as the ensemble in [44] at a substantially reduced computational cost. Note that we did not preprocess the faces and we used the automatically detected and aligned faces provided in the SFEW dataset.

Table IV shows the results of the proposed SJMT approach compared with the classical multitask approach and the individual networks using the Oulu-CASIA dataset for emotion

TABLE V

COMPARISON WITH STATE-OF-THE-ART SOLUTIONS ON THE OULU-CASIA DATASET. (*) INDICATES APPROACHES USING SEQUENCES AND NOT ONLY STATIC IMAGES

Method	Oulu-CASIA
Kacem et al. [47]*	83.1%
Ding et al. [46]	82.2%
Liu et al. [48] *	74.6%
Zhao et al. [49] *	84.6%
Jung et al. [50]*	81.4%
SJMT	85.1%

recognition and the EmotioNet dataset for AU detection. As with the SFEW dataset, both multitask approaches outperformed the single networks and SJMT obtained the best accuracy for emotion recognition: 85.1% with a VGG-16 architecture. However, in this experiment, for AU detection, the best accuracy was achieved by the classical multitask approach (88.2%). In terms of sensitivity, the same situation as with the SFEW dataset occurred: the individual networks obtained better performance for that measure than the multitask approaches.

Table V reports the results on the Oulu-CASIA dataset. Since this dataset is composed of videos, the methods using a sequence of frames are denoted with *. In this case, the proposed SJMT method achieved the best results in comparison to the existing methods in the state of the art (+0.5%).

C. Qualitative Results for SJMT

To analyze the benefits of using the proposed SJMT approach with related tasks, we focused on the predictions for AUs when feeding the model with images from the SFEW and Oulu-CASIA datasets, in which only labels for emotion recognition are provided. Thus, we grouped the images by the predicted emotion category and computed the mean scores for each AU as shown in Fig. 2. This figure shows that when the model recognizes an emotion category, it is simultaneously capable to identify the corresponding AUs in most of the cases. In order to simplify the interpretation of the figure, Table VI shows the description of the AUs predicted by SJMT along with each emotion and a representative image from the test set of the SFEW and Oulu-CASIA datasets.

The table shows the coherence of the approach when recognizing the AUs in emotion-labeled images. Given the fact that only 11 AU labels are used in the EmotioNet dataset (which is the dataset used to train the SJMT approach), some of the AUs theoretically forming the emotions cannot be recognized. A clear example is the *Sad* emotion: in theory, the most discriminant AU is 15 (lip corner depressor), but this AU is not present in the dataset. Thus, the model predicts as most discriminant AU4 (brow lowerer) for SFEW and AU17 (chin raiser) for Oulu-CASIA images. Furthermore, some differences in the predictions of AUs can be seen between both datasets. These differences can be explained by the way the emotions are expressed by the subjects. In SFEW, facial expressions are performed by professional actors mimicking real-life emotion expressions, while in the Oulu-CASIA dataset participants performed a more stereotypical emotion expression.

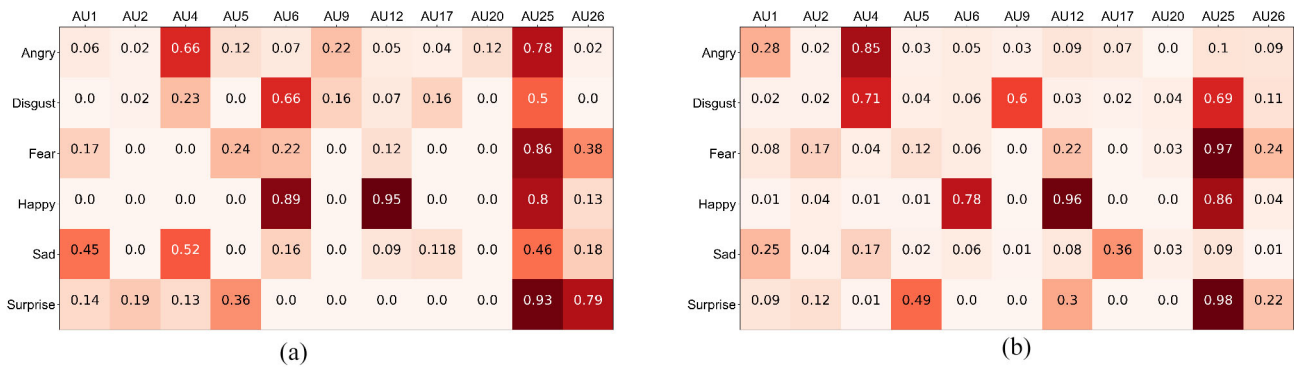
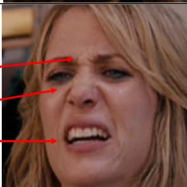
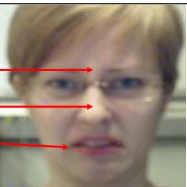
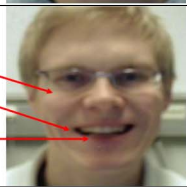
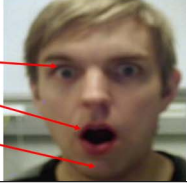


Fig. 2. Mean scores for AUS where only emotion labels are provided using the SJMT approach for (a) SFEW dataset and (b) Oulu-CASIA dataset.

TABLE VI
DESCRIPTION OF THE RECOGNIZED AUS FOR EACH EMOTION

Emotion	Examples of AUS recognized in the SFEW dataset	Examples of AUS recognized in the Oulu-CASIA dataset
Angry	 AU 4: Brow Lowerer AU 9: Nose Wrinkler AU 25: Lips Part	 AU 1: Inner Brow Raiser AU 4: Brow Lowerer
Disgust	 AU 4: Brow Lowerer AU 6: Cheek Raiser AU 25: Lips Part	 AU 4: Brow Lowerer AU 9: Nose Wrinkler AU 25: Lips Part
Fear	 AU 5: Upper Lid Raiser AU 25: Lips Part AU 26: Jaw Drop	 AU 25: Lips Part AU 12: Lip Corner Puller AU 26: Jaw Drop
Happy	 AU 6: Cheek Raiser AU 12: Lip Corner Puller AU 25: Lips Part	 AU 6: Cheek Raiser AU 12: Lip Corner Puller AU 25: Lips Part
Sad	 AU 1: Inner Brow Raiser AU 4: Brow Lowerer AU 25: Lips Part	 AU 1: Inner Brow Raiser AU 17: Chin Raiser
Surprise	 AU 5: Upper Lid Raiser AU 25: Lips Part AU 26: Jaw Drop	 AU 5: Upper Lid Raiser AU 25: Lips Part AU 26: Jaw Drop

D. Compound Emotion Recognition

Most of the research on emotion recognition has focused on the study of the six basic categories. We use the term “basic” to refer to the fact that such emotion categories cannot be decomposed into smaller semantic labels. However, it has

been shown [51] that many more facial expressions of emotion exist and are used regularly by humans. Compound emotions are those that can be constructed by combining basic component categories to create new ones. Hence, in this section, we will validate our method on the extended problem

TABLE VII
RESULTS OF SINGLE AND MULTITASK LEARNING FOR COMPOUND
EMOTION RECOGNITION TASKS IN THE PERCENTAGE OF ACCURACY

Compound emotion	Number images	Single Resnet-50	SJMT Resnet-50
Angrily disgusted	19	35.0%	52.0%
Angrily surprised	25	36.0%	60.0%
Fearfully angry	19	42.1%	52.6%
Fearfully surprised	17	76.4%	76.5%
Happily disgusted	486	57.4%	92.6%
Happily surprised	36	88.8%	77.8%
Sadly angry	15	66.7%	53.3%
Sadly disgusted	105	33.3%	70.5%
Mean acc. compounds	-	54.5%	66.9%
Acc. all images	722	54.3%	84.2%

of recognizing the following compound emotions from the EmotionNet dataset [37]: *angrily disgusted*, *angrily surprised*, *fearfully angry*, *fearfully surprised*, *happily disgusted*, *happily surprised*, *sadly angry*, and *sadly disgusted*. The number of available samples is scarce for most of the classes and we hypothesize that this emotion prediction task can take benefit from a joint learning process along with an auxiliary task such as AU recognition.

Following the experiments of the previous sections, in Table VII, we compare the performance of the SJMT approach for compound emotion images with a single dedicated task network, in both cases, training a Resnet-50. Given the unbalanced number of images per class, we decided to train the networks with a set of 160 images consisting of 20 images in each class. During the training of the SJMT approach, the remaining EmotionNet images that only contain AU labels are used to train the AU recognition task.

Table VII shows the accuracy obtained for each compound emotion for both single network and SJMT methods. *Acc. all images* is the accuracy for all the images in the dataset. *Mean acc. compounds* refers to the mean value of the accuracies obtained for each compound emotion. This measure gives the same weight to all the emotions without considering the number of images per class. Regarding the results of each emotion, the SJMT improves the accuracy in 6 out of 8 of them. The SJMT also considerably improves the accuracy when considering all the images of the dataset, from 54.3% to 84.2%. This improvement can be explained by the improvement in the recognition of the two classes with the largest number of images, *happily disgusted* (composed of 486 images, from 57.4% to 92.6%) and *sadly disgusted* (composed of 105 images from 33.3% to 70.5%). Not taking into account the differences in the number of images by computing the mean value of each emotion accuracy, the SJMT also improves the single network performance (+12.4%).

V. CONCLUSION

In this article, we presented the *selective joint multi-task* approach, which defines a novel datasetwise selective sigmoid cross-entropy loss function to address multitask, multilabel, and multidataset problems. Specifically, this proposal addresses one of the challenges with discrete emotion recognition *in the wild*, which is the lack of large public

labeled datasets. By including a joint training with a large dataset for AU recognition in a multitask approach, the results for emotion recognition were improved.

Our proposed approach was assessed using the SFEW and Oulu-CASIA datasets for emotion recognition and the EmotionNet dataset for AU detection, and the results were compared to single task networks dedicated to each task individually and to the classical multitask approach. SJMT obtained the best results in terms of accuracy for the emotion recognition task in all the experiments. Results also showed benefits of learning multiple correlated tasks simultaneously by demonstrating visually that even for images without AU labels, the model is capable of inferring them coherently. Finally, we evaluated the proposed method in a dataset of compound emotions, again obtaining improved results.

Future work will consist of studying and developing the inclusion of the landmark detection task in this multitask approach. We believe that the addition of structural and geometrical cues to the appearance-based CNN might boost the performance of the emotion recognition task, but a new reformulation of the selective cross-entropy function needs to be done to handle such a problem in joint training.

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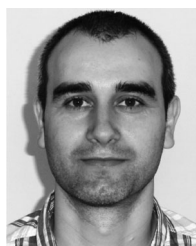
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Gerard Pons received the graduation degree in computer science engineering, the master's degree in industrial, automation, and computer systems, and the Ph.D. degree from the University of Girona, Girona, Spain, in 2009, 2010, and 2014, respectively.

He is a Postdoctoral Researcher with the Centrum Wiskunde and Informatica, Amsterdam, The Netherlands. His master thesis was done in collaboration with the Institute of Biomedical Engineering, University of Oxford, Oxford, U.K.

Both Master and Ph.D. theses were conducted in breast imaging, particularly in detection and segmentation of breast tumors in ultrasonography. He was part of the Universitat Oberta de Catalunya, Barcelona, Spain, until 2019, when his main research topic changed to affective computing and emotion recognition. Collaborating in research since 2007, his main research interests include computer vision, image processing, artificial intelligence, pattern recognition, and machine learning.



David Masip (Senior Member, IEEE) received the graduation degree in computer science from the Universitat Autònoma de Barcelona, Bellaterra, Spain, in 2001, and the Ph.D. degree from the Computer Vision Center, Barcelona, Spain, in September 2005.

He has been an Associate Professor with the Computer Science, Multimedia and Telecommunications Department, Universitat Oberta de Catalunya, Barcelona, since 2007, where he has been the Director of the UOC Doctoral School since 2015. He is the Director of the Scene Understanding and Artificial Intelligence Research Group. Previously, he worked as an Assistant Professor with the Applied Mathematics Department, Universitat de Barcelona, Barcelona. His main research interests are computer vision, deep learning algorithms, and facial expression classification.

Dr. Masip received the Best Thesis Award on Computer Science in the Universitat Autònoma de Barcelona.